

Neil Farbotnik

SER 321

Assn 2

1. <https://api.github.com/repos/AwangWOW/Wangbar/commits>

```
api.github.com/repos/AwangWOW/Wangbar/commits

{
  "sha": "e3b02f69e596702f308dbbfc4c64a201f1edbda",
  "node_id": "C_kwDORB3x6doAGUzYjYyZjY5ZTU5NjcwMmYzMDhiZGJiZWw0YzY0YTIwMmYxZWw0ZGZGE",
  "commit": {
    "author": {
      "name": "AwangWOW",
      "email": "109743204+AwangWOW@users.noreply.github.com",
      "date": "2026-02-09T01:42:00Z"
    },
    "committer": {
      "name": "GitHub",
      "email": "noreply@github.com",
      "date": "2026-02-09T01:42:00Z"
    },
    "message": "Merge pull request #24 from zombfeed/hideoutofcombat\\n\\n+ fix energy not reappearing when disabling hide options",
    "tree": {
      "sha": "4d4d2aedd19a6a0962516d5dfd5f30d4381e99a1",
      "url": "https://api.github.com/repos/AwangWOW/Wangbar/git/trees/4d4d2aedd19a6a0962516d5dfd5f30d4381e99a1"
    },
    "url": "https://api.github.com/repos/AwangWOW/Wangbar/git/commits/e3b02f69e596702f308dbbfc4c64a201f1edbda",
    "comment_count": 0,
    "verification": {
      "verified": true,
      "reason": "valid",
      "signature": "-----BEGIN PGP SIGNATURE-----\n/nwsFcBAACAAQ8QJpIToCRciaQ7uu5UhlAAgDkQAIZDOHwU5VmU0gtUfWp75gC\\ne3s0ld4UKMKG35J53uq2QcGPrZ0ZK7zaHZEw6ulCE+91v2wQp8qN9IIhiZiPhXoh\\nTgDNkBXTRPjIys9LWFH\n5RjgPaxSlagKa+It4oM0/pWEHVe/0I4ioECeR9GwuhwRHXVyaR6Cf\\nxvFvVj900zY+rzavv7yN+pmTKvVwOVVxN1cD2VVP4KInFYBPQKMMg/fQb+QfEI\\n9kms5Bf-jwXIV3TEIYSNZcB6+qcdzDYLVj\nKpGAGjg5Quxh2Y6JKj7BhWLGWk59cmMLZhO36F74\\nGx0XwL+Lk0zxinIbdtXOggKh4yS0Nvo3+R2DzgUoINBzP2LmccUDtASnVM8K34H\\nswktDfVsm2tC5LPhJqo7vmacEbMGRpS6AvykGwErQHmMzG\nLnkH4Iou\\nmcDmp+nCVbHYSdY\\nbz:8EAJ81VS7X7s:chGrx6i358Lt0ZqW60hKMSyblYnXU7VGAJt1x+3MbVYK13ZAU1\\nbaz9CtuTlNyU3er3j8a\\n=zm09\\n-----END PGP SIGNATURE-----\\n",
      "payload": "tree 4d4d2aedd19a6a0962516d5dfd5f30d4381e99a1\\nparent 6b3d7b166ba0a73473e99d3bd84234a787a79473\\nparent aebb14c1cf7f3d5b1c792cea5744ceb4\n109743204+AwangWOW@users.noreply.github.com 1770601320 -0600\\ncommitter GitHub <noreply@github.com> 1770601320 -0600\\n\\nMerge pull request #24 from zombf\nisabling hide options",
      "verified_at": "2026-02-09T01:42:01Z"
    }
  },
  "url": "https://api.github.com/repos/AwangWOW/Wangbar/commits/e3b02f69e596702f308dbbfc4c64a201f1edbda",
  "html_url": "https://github.com/AwangWOW/Wangbar/commit/e3b02f69e596702f308dbbfc4c64a201f1edbda",
  "comments_url": "https://api.github.com/repos/AwangWOW/Wangbar/commits/e3b02f69e596702f308dbbfc4c64a201f1edbda/comments",
  "author": {
    "login": "AwangWOW",
    "id": 109743204,
    "node_id": "U_kgDOB0qMZA",
    "avatar_url": "https://avatars.githubusercontent.com/u/109743204?v=4",
    "gravatar_id": "",
    "url": "https://api.github.com/users/AwangWOW",
    "html_url": "https://github.com/AwangWOW",
    "followers_url": "https://api.github.com/users/AwangWOW/followers",
    "following_url": "https://api.github.com/users/AwangWOW/following{/other_user}",
    "gists_url": "https://api.github.com/users/AwangWOW/gists{/gist_id}",
    "starred_url": "https://api.github.com/users/AwangWOW/starred{/owner}/{/repo}",
    "subscriptions_url": "https://api.github.com/users/AwangWOW/subscriptions",
    "organizations_url": "https://api.github.com/users/AwangWOW/orgs",
    "repos_url": "https://api.github.com/users/AwangWOW/repos",
    "events_url": "https://api.github.com/users/AwangWOW/events{/privacy}",
    "received_events_url": "https://api.github.com/users/AwangWOW/received_events",
    "type": "User",
    "user_view_type": "public",
    "site_admin": false
  },
  "committer": {
    "login": "web-flow",
    "id": 19864447,
    "node_id": "MDQ6VXNlcjE5ODY0NDQ3",
    "avatar_url": "https://avatars.githubusercontent.com/u/19864447?v=4",
    "gravatar_id": "",
    "url": "https://api.github.com/users/web-flow",
    "html_url": "https://github.com/web-flow",
    "followers_url": "https://api.github.com/users/web-flow/followers",
    "following_url": "https://api.github.com/users/web-flow/following{/other_user}",
    "gists_url": "https://api.github.com/users/web-flow/gists{/gist_id}",
    "starred_url": "https://api.github.com/users/web-flow/starred{/owner}/{/repo}",
    "subscriptions_url": "https://api.github.com/users/web-flow/subscriptions",
    "organizations_url": "https://api.github.com/users/web-flow/orgs",
    "repos_url": "https://api.github.com/users/web-flow/repos",
    "events_url": "https://api.github.com/users/web-flow/events{/privacy}",
    "received_events_url": "https://api.github.com/users/web-flow/received_events",
    "type": "User",
    "user_view_type": "public",
    "site_admin": false
  }
}
```

2. https://api.github.com/repos/AwangWOW/Wangbar/commits?sha=bigupdate1&per_page=40

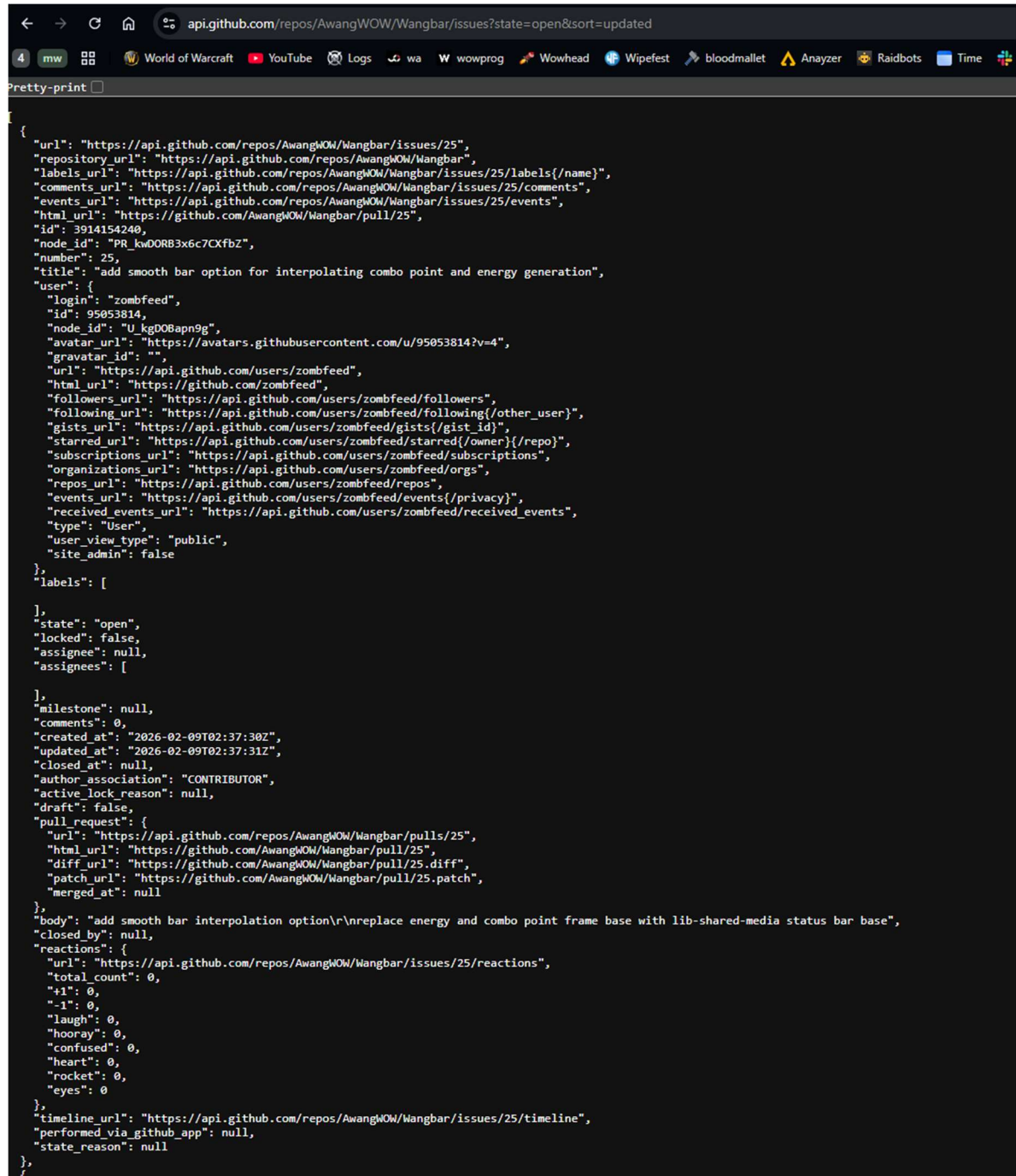
```
api.github.com/repos/AwangWOW/Wangbar/commits?sha=bigupdate1&per_page=40

4 mw World of Warcraft YouTube Logs wa W wowprog Wowhead Wipefest bloodmallet Anayzer Raid

Pretty-print [ ]

[
  {
    "sha": "a40644114ff795db65d576b9c4e73cc91cf65210",
    "node_id": "C_kwDORB3x6doAKGE0MDY0NDEExNGZmNzk1ZGI2NWQ1NzZiOWM0ZTczY2M5MlNjUyMTA",
    "commit": {
      "author": {
        "name": "AwangWOW",
        "email": "aaronmaslowski9@gmail.com",
        "date": "2026-02-01T23:30:34Z"
      },
      "committer": {
        "name": "AwangWOW",
        "email": "aaronmaslowski9@gmail.com",
        "date": "2026-02-01T23:30:34Z"
      },
      "message": "arcui fix",
      "tree": {
        "sha": "a79f9b800e760476cfe9bc2a0c62660e7a05f327",
        "url": "https://api.github.com/repos/AwangWOW/Wangbar/git/trees/a79f9b800e760476cfe9bc2a0c62660e7a05f327"
      },
      "url": "https://api.github.com/repos/AwangWOW/Wangbar/git/commits/a40644114ff795db65d576b9c4e73cc91cf65210",
      "comment_count": 0,
      "verification": {
        "verified": false,
        "reason": "unsigned",
        "signature": null,
        "payload": null,
        "verified_at": null
      }
    },
    "url": "https://api.github.com/repos/AwangWOW/Wangbar/commits/a40644114ff795db65d576b9c4e73cc91cf65210",
    "html_url": "https://github.com/AwangWOW/Wangbar/commit/a40644114ff795db65d576b9c4e73cc91cf65210",
    "comments_url": "https://api.github.com/repos/AwangWOW/Wangbar/commits/a40644114ff795db65d576b9c4e73cc91cf65210/comments",
    "author": {
      "login": "AwangWOW",
      "id": 109743204,
      "node_id": "U_kgDOBoqMZA",
      "avatar_url": "https://avatars.githubusercontent.com/u/109743204?v=4",
      "gravatar_id": "",
      "url": "https://api.github.com/users/AwangWOW",
      "html_url": "https://github.com/AwangWOW",
      "followers_url": "https://api.github.com/users/AwangWOW/followers",
      "following_url": "https://api.github.com/users/AwangWOW/following{/other_user}",
      "gists_url": "https://api.github.com/users/AwangWOW/gists{/gist_id}",
      "starred_url": "https://api.github.com/users/AwangWOW/starred{/owner}/{/repo}",
      "subscriptions_url": "https://api.github.com/users/AwangWOW/subscriptions",
      "organizations_url": "https://api.github.com/users/AwangWOW/orgs",
      "repos_url": "https://api.github.com/users/AwangWOW/repos",
      "events_url": "https://api.github.com/users/AwangWOW/events{/privacy}",
      "received_events_url": "https://api.github.com/users/AwangWOW/received_events",
      "type": "User",
      "user_view_type": "public",
      "site_admin": false
    },
    "committer": {
      "login": "AwangWOW",
```

3. <https://api.github.com/repos/AwangWOW/Wangbar/issues?state=open&sort=updated>



```
{
  "url": "https://api.github.com/repos/AwangWOW/Wangbar/issues/25",
  "repository_url": "https://api.github.com/repos/AwangWOW/Wangbar",
  "labels_url": "https://api.github.com/repos/AwangWOW/Wangbar/issues/25/labels{/name}",
  "comments_url": "https://api.github.com/repos/AwangWOW/Wangbar/issues/25/comments",
  "events_url": "https://api.github.com/repos/AwangWOW/Wangbar/issues/25/events",
  "html_url": "https://github.com/AwangWOW/Wangbar/pull/25",
  "id": 3914154240,
  "node_id": "PR_kwDORB3x6c7CXfbZ",
  "number": 25,
  "title": "add smooth bar option for interpolating combo point and energy generation",
  "user": {
    "login": "zombfeed",
    "id": 95053814,
    "node_id": "U_kgD0Bapn9g",
    "avatar_url": "https://avatars.githubusercontent.com/u/95053814?v=4",
    "gravatar_id": "",
    "url": "https://api.github.com/users/zombfeed",
    "html_url": "https://github.com/zombfeed",
    "followers_url": "https://api.github.com/users/zombfeed/followers",
    "following_url": "https://api.github.com/users/zombfeed/following{/other_user}",
    "gists_url": "https://api.github.com/users/zombfeed/gists{/gist_id}",
    "starred_url": "https://api.github.com/users/zombfeed/starred{/owner}/{/repo}",
    "subscriptions_url": "https://api.github.com/users/zombfeed/subscriptions",
    "organizations_url": "https://api.github.com/users/zombfeed/orgs",
    "repos_url": "https://api.github.com/users/zombfeed/repos",
    "events_url": "https://api.github.com/users/zombfeed/events{/privacy}",
    "received_events_url": "https://api.github.com/users/zombfeed/received_events",
    "type": "User",
    "user_view_type": "public",
    "site_admin": false
  },
  "labels": [
  ],
  "state": "open",
  "locked": false,
  "assignee": null,
  "assignees": [
  ],
  "milestone": null,
  "comments": 0,
  "created_at": "2026-02-09T02:37:30Z",
  "updated_at": "2026-02-09T02:37:31Z",
  "closed_at": null,
  "author_association": "CONTRIBUTOR",
  "active_lock_reason": null,
  "draft": false,
  "pull_request": {
    "url": "https://api.github.com/repos/AwangWOW/Wangbar/pulls/25",
    "html_url": "https://github.com/AwangWOW/Wangbar/pull/25",
    "diff_url": "https://github.com/AwangWOW/Wangbar/pull/25.diff",
    "patch_url": "https://github.com/AwangWOW/Wangbar/pull/25.patch",
    "merged_at": null
  },
  "body": "add smooth bar interpolation option\r\n\r\nreplace energy and combo point frame base with lib-shared-media status bar base",
  "closed_by": null,
  "reactions": {
    "url": "https://api.github.com/repos/AwangWOW/Wangbar/issues/25/reactions",
    "total_count": 0,
    "+1": 0,
    "-1": 0,
    "laugh": 0,
    "hooray": 0,
    "confused": 0,
    "heart": 0,
    "rocket": 0,
    "eyes": 0
  },
  "timeline_url": "https://api.github.com/repos/AwangWOW/Wangbar/issues/25/timeline",
  "performed_via_github_app": null,
  "state_reason": null
},
{
}
```

1. The first request, <https://api.github.com/repos/AwangWOW/Wangbar/commits>, calls GET /repos/{owner}/{repo}/commits and returns commits from the repository's default branch. It required the repository owner (AwangWOW) and repository name (Wangbar) in the URL path, and since no query parameters were included, GitHub automatically returned commits from the default branch.

<https://docs.github.com/en/rest/commits/commits>

The second request,

https://api.github.com/repos/AwangWOW/Wangbar/commits?sha=bigupdate1&per_page=40, uses the same endpoint but adds query parameters to the HTTP request.

The parameter sha=bigupdate1 specifies a non-default branch, and per_page=40 increases the number of commits returned in the response. These parameters are included in the URL and modify what the server returns. Documentation:

<https://docs.github.com/en/rest/commits/commits#list-commits>

The third request,

<https://api.github.com/repos/AwangWOW/Wangbar/issues?state=open&sort=updated>, calls /repos/{owner}/{repo}/issues to retrieve repository issues. The state=open parameter filters for open issues, and sort=updated orders them by most recently updated. Documentation: <https://docs.github.com/en/rest/issues/issues#list-repository-issues>

2. HTTP communication is an example of being stateless, which means that each request is treated as completely independent. The server does not store information about previous requests. In the examples above, each API call was processed separately by GitHub. The server did not rely on information from earlier requests when handling later ones. Every request had to include all required information, like the repository name and query parameters. Stateful communication is different because the server keeps track of prior interactions. An example being when a user logs into GitHub in a browser and remains logged in while navigating between repositories, issues, and pull requests.

Terminal Output:

```

No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
  README.md
nothing added to commit but untracked files present (use "git add" to track)
[ec2-user@ip-172-31-14-155 asan2]$ cd ser321-spring26-C-campus/
[ec2-user@ip-172-31-14-155 ser321-spring26-C-campus]$ ls
Assignment1 Assignment2 Assignment3 Assignment4 Assignment5 Assignment6 README.md
[ec2-user@ip-172-31-14-155 ser321-spring26-C-campus]$ cd Assignment2
[ec2-user@ip-172-31-14-155 Assignment2]$ ls
WebServer
[ec2-user@ip-172-31-14-155 Assignment2]$ cd WebServer/
[ec2-user@ip-172-31-14-155 WebServer]$ ls
README.md build.gradle src www
[ec2-user@ip-172-31-14-155 WebServer]$ gradle FunWebServer
Starting a Gradle Daemon (subsequent builds will be faster)
> Task :compileJava
Note: /home/ec2-user/asan2/ser321-spring26-C-campus/Assignment2/WebServer/src/main/java/fun/HttpServer/WebServer.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
> Task :FunWebServer
Received: GET / HTTP/1.1
Received: Host: 3.148.182.86:9000
Received: Connection: keep-alive
Received: Upgrade-Insecure-Requests: 1
Received: User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36
Received: Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Received: Accept-Encoding: gzip, deflate
Received: Accept-Language: en-US,en;q=0.9,vi-VN;q=0.8,vi;q=0.7,en-CA;q=0.6,ru-RU;q=0.5,ru;q=0.4,ko-KR;q=0.3,ko;q=0.2
Received:
FINISHED PARING HEADER
-----> 75% EXECUTING (1m 37s)
> :FunWebServer

```

Web Browser (Net secure 3.148.182.86:9000):

You can make the following GET requests

- /file/sample.html -- returns the content of the file sample.html
- /json -- returns a json of the /random request
- /random -- returns index.html

File Structure in www (you can use /file/www/FILENAME):

- index.html
- root.html

i-00b313720de8ec88 (SER321)
PublicPc: 3.148.182.86 PrivatePc: 172.31.14.155

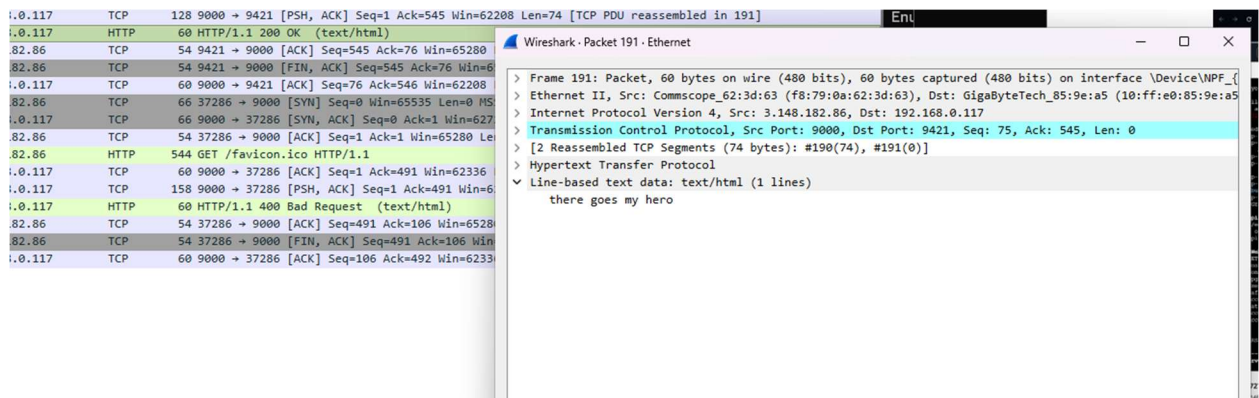
2.2

tcp.port==9000									
No.	Time	Source	Destination	Protocol	Leng	Info			
148	8.401483	192.168.0.117	3.148.182.86	TCP	66	9421 → 9000 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM			
179	8.474824	3.148.182.86	192.168.0.117	TCP	66	9000 → 9421 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=128			
180	8.474861	192.168.0.117	3.148.182.86	TCP	54	9421 → 9000 [ACK] Seq=1 Ack=1 Win=65280 Len=0			
181	8.474965	192.168.0.117	3.148.182.86	HTTP	598	GET /addline?text=there%20goes%20my%20hero HTTP/1.1			
189	8.551136	3.148.182.86	192.168.0.117	TCP	60	9000 → 9421 [ACK] Seq=1 Ack=545 Win=62208 Len=0			
190	8.552738	3.148.182.86	192.168.0.117	TCP	128	9000 → 9421 [PSH, ACK] Seq=1 Ack=545 Win=62208 Len=74 [TCP PDU reassembled in 191]			
191	8.552836	3.148.182.86	192.168.0.117	HTTP	60	HTTP/1.1 200 OK (text/html)			
192	8.552852	192.168.0.117	3.148.182.86	TCP	54	9421 → 9000 [ACK] Seq=545 Ack=76 Win=65280 Len=0			
193	8.553414	192.168.0.117	3.148.182.86	TCP	54	9421 → 9000 [FIN, ACK] Seq=545 Ack=76 Win=65280 Len=0			
201	8.626961	3.148.182.86	192.168.0.117	TCP	60	9000 → 9421 [ACK] Seq=76 Ack=546 Win=62208 Len=0			
261	8.928350	192.168.0.117	3.148.182.86	TCP	66	37286 → 9000 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM			
264	9.004943	3.148.182.86	192.168.0.117	TCP	66	9000 → 37286 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=128			
265	9.004986	192.168.0.117	3.148.182.86	TCP	54	37286 → 9000 [ACK] Seq=1 Ack=1 Win=65280 Len=0			
266	9.005098	192.168.0.117	3.148.182.86	HTTP	544	GET /favicon.ico HTTP/1.1			
268	9.008942	3.148.182.86	192.168.0.117	TCP	60	9000 → 37286 [ACK] Seq=1 Ack=491 Win=62336 Len=0			
269	9.082507	3.148.182.86	192.168.0.117	TCP	158	9000 → 37286 [PSH, ACK] Seq=1 Ack=491 Win=62336 Len=104 [TCP PDU reassembled in 270]			
270	9.082688	3.148.182.86	192.168.0.117	HTTP	60	HTTP/1.1 400 Bad Request (text/html)			
271	9.082707	192.168.0.117	3.148.182.86	TCP	54	37286 → 9000 [ACK] Seq=491 Ack=106 Win=65280 Len=0			
272	9.083229	192.168.0.117	3.148.182.86	TCP	54	37286 → 9000 [FIN, ACK] Seq=491 Ack=106 Win=65280 Len=0			
274	9.156996	3.148.182.86	192.168.0.117	TCP	60	9000 → 37286 [ACK] Seq=106 Ack=492 Win=62336 Len=0			

2.3

1. I chose tcp.port == 9000, this is because I wanted to capture all tcp traffic to the server.
2. I am able to receive http codes like 200, 400, 404, 500, etc.
3. The server returns 200 OK when a valid request like is run. It also returns 400 bad request for a bad request, both of these are shown in green.

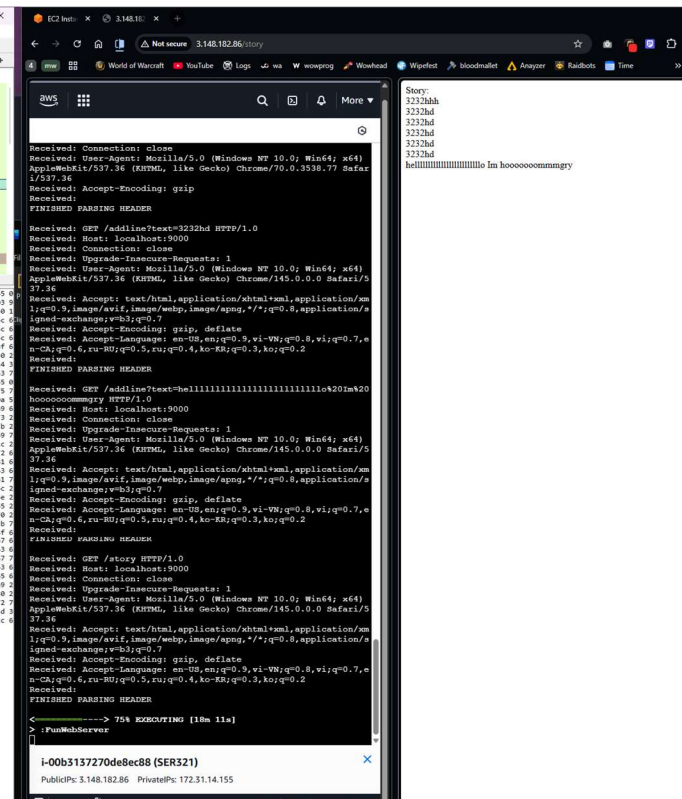
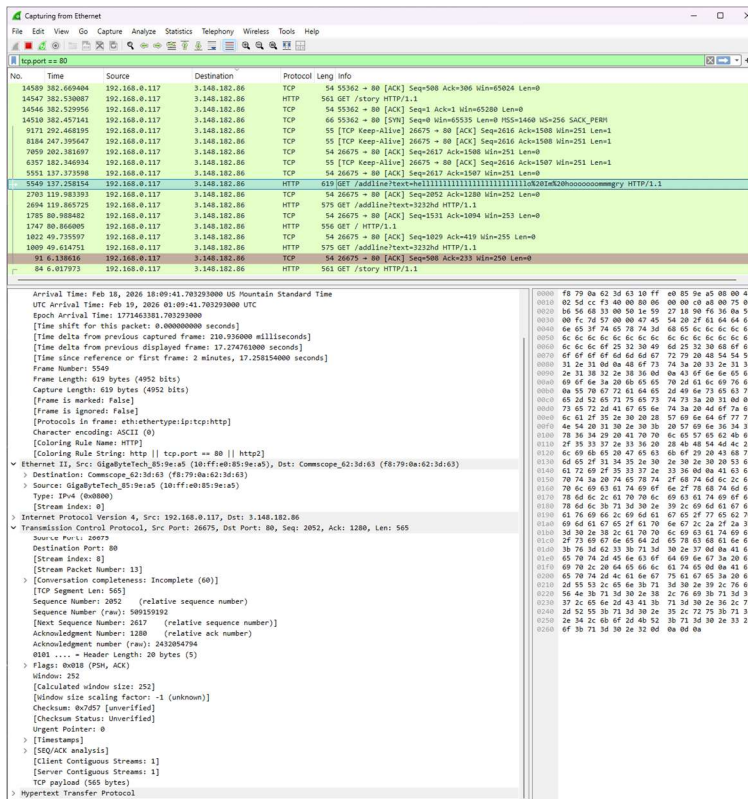
4. Yes I was able to find the data.



5. HTTPS encrypts the data so any bozo off the street with a laptop cant just sniff all the data being transferred on a network and extract useful information like credit card numbers, names, etc. Its more common as the added security is worth it in almost every circumstance.
6. Port 9000. No its not the most common, its just what we chose. Port 80 and 443 are more common in real applications.
7. The local port used to send requests to the Web Server. The server's port remains 9000, but the client port is different each time since its randomly assigned.

2.4

- 1) 3.148.182.86 (ipV4 from aws)
- 2) It is different. Its now port 80
- 3) HTTP. The text is not exrypted. And typing the IP v4 addresss is enough to access it. I don't need HTTPS prefix.
- 4) Yes in theory I could change my AWS settings to modify the security but I changing the AWS setting alone are not enough to go to HTTPS.
- 5) Below



2.5.1

I added error handling to the multiply? endpoint to prevent the server from crashing when invalid or missing input is provided. First, I added checks to handle cases where num1= or num2= are missing from the request. If either parameter is not included, the server now returns a 400 Bad Request response instead of failing. I also added a check whether the query_pairs given are null or empty strings. If either value is missing or blank, the server responds with an HTTP error message and displays an error on the screen, but it does not crash. Additionally, I changed the parameter handling so that num1 and num2 are first treated as strings instead of immediately being parsed as integers. I then placed the multiplication logic inside a try-catch statement. If parsing fails and a NumberFormatException is thrown, the catch block returns a 400 Bad Request response and informs the user that the submitted values are not valid integers. This guarantees that incorrect input does not terminate the server and that proper HTTP error codes are returned.

